

DOCUMENT 1 OF 3

Build and technology

How the Metal Prod Service website was built, which technologies were chosen, and why.

1. Purpose

The site is the public face of Metal Prod Service S.R.L. (Parc Industrial Sud 2/B, Satu Mare, CUI RO18205726). It must present CNC milling, turning, prototyping, series work, technical plastics and in-house heat treatment, in seven languages, and collect a technical request-for-quote (RFQ) that engineering can act on within 24 hours.

2. Architecture

A static single-page application (SPA). There is no application server and no database. Vite compiles TypeScript and CSS into hashed assets under **dist/**. Preview and production are just a static file host (Vite preview locally; any HTTPS host in production) plus security headers.

Layer	Choice	Role
Language	TypeScript (strict)	UI, RFQ, i18n, security helpers
Bundler	Vite 6	Dev server, production build, header injection
3D	Three.js 0.170	Hero CNC spur gear (lazy-loaded chunk)
Markup	index.html	Semantic sections, RFQ form, JSON-LD
Style	Custom CSS tokens	Industrial palette from the specification, no Tailwind
i18n	data-i18n dictionaries	EN, RO, HU, DE, FR, ES, IT
Tests	Vitest + Playwright	Unit, e2e, foldable viewports
Load	Node fetch probe	80 concurrent users × 4 rounds
RFQ delivery	FormSubmit HTTPS + mailto fallback	Owner inbox trif_marius1@yahoo.com

3. Information architecture

- Header: logo, primary nav (desktop), language, theme, RFQ, burger (≤980 px).
- Hero: factory photo, ISO chips, headline, two CTAs, 3D gear.
- Trust strip, About (factory exterior + interior photos), industries, six services, process, catalogue, equipment table, RFQ, FAQ, contact + OSM map, footer, mobile bottom nav.

4. RFQ engine

Three steps: (1) production mode, material, heat treatment; (2) CAD/drawing upload with magic-byte checks; (3) company, engineer, email, phone, date, notes, GDPR. Submit posts JSON to FormSubmit (owner Yahoo inbox) and falls back to a pre-filled mailto if the HTTP path is not yet activated. Files are vetted in the browser (extension allow-list, blocked executables, 50 MB cap, 8 files, PDF/ZIP/STEP/IGES/DXF/DWG signatures). A honeypot plus a 1.4 s minimum fill time reject obvious bots. Control characters and mailto header injection are stripped.

5. Security posture (static site)

- CSP: script-src 'self' (no unsafe-eval / unsafe-inline), object-src none, worker-src none, upgrade-insecure-requests, connect-src self + formsubmit.co.
- COOP / CORP same-origin, X-Content-Type-Options nosniff, X-Frame-Options SAMEORIGIN, HSTS, tight Permissions-Policy.

- No PII in sessionStorage. File names HTML-escaped. OSM iframe sandboxed.

6. Responsive and foldable design

Breakpoints treat cover screens as phones, unfolded book-style inners (~600–840 px, near-square) as small tablets (two-column hero and services), Flip covers and flex/tent panes as short one-column views, and landscape phones as two-column heroes without a bottom nav. CSS Viewport Segments pad dual-screen hinges when the browser exposes them.

7. Source layout

Path	Contents
src/main.ts	Boot, i18n apply, header, RFQ wiring, 3D lazy import
src/rfq.ts	State, validation, mailto, FormSubmit payload
src/security.ts	Magic bytes, sanitise, email/phone/name rules
src/logo3d.ts	CNC spur gear scene
src/i18n*.ts	Seven locale dictionaries
src/styles.css	Tokens, layout, foldable media queries
tests/	Vitest, Playwright e2e, load probe, visual scripts
public/	_headers, robots, sitemap, factory and product images

8. What is intentionally not in this build

No CMS, no customer login, no server-side CAD virus scan or ZIP unpack, no native iOS/Android apps. Those belong in a later backend. The current site is the marketing and RFQ front door.